

Fields and Waves

Analytical Modeling and 3D Full-Wave Simulation of RG-187A/U Coaxial Cable

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Abstract

This report investigates the high-frequency transmission characteristics of an RG-187A/U coaxial cable operating at 1 GHz. Distributed per-unit-length parameters (R, L, C, G) are calculated theoretically using MATLAB. These analytical outcomes are compared against 3D full-wave electromagnetic simulations performed in CST Studio Suite, followed by a reverse-calculation to extract effective inductance and capacitance from numerical S-parameter outputs.

1 Analytical Formulation (MATLAB)

The coaxial cable consists of an inner conductor with radius $a = 0.3$ mm, an outer conductor with inner radius $b = 1.6$ mm, and a PTFE dielectric core characterized by $\epsilon_r = 2.3$. At an operating frequency of $f = 1$ GHz, the skin depth (δ) confines the current to the conductor surfaces.

The distributed parameters are analytically determined as:

$$R = \frac{1}{2\pi\delta\sigma_c} \left(\frac{1}{a} + \frac{1}{b} \right) \quad (1)$$

$$L = \frac{\mu_0}{2\pi} \ln \left(\frac{b}{a} \right) \quad (2)$$

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln(b/a)} \quad (3)$$

$$G = \frac{2\pi\sigma_d}{\ln(b/a)} \approx 0 \text{ (Lossless dielectric)} \quad (4)$$

Implementing these formulas in MATLAB yields a theoretical characteristic impedance of $Z_0 = 66.18 \Omega$ and a phase constant of $\beta = 31.78$ rad/m.

2 CST Studio Suite Verification

A 3D model of the coaxial waveguide was constructed in CST Studio Suite. Due to numerical discretization and boundary conditions, slight deviations between full-wave simulation and 1D analytical theory are expected. Table 1 summarizes the results.



Table 1: Comparison of Transmission Parameters at 1 GHz

Parameter	MATLAB (Theory)	CST (Simulation)
Characteristic Impedance (Z_0)	66.18 Ω	69.2213 Ω
Phase Constant (β)	31.78 rad/m	15.18 rad/m

3 Extraction of Distributed Parameters from CST Data

Using the fundamental transmission line equations for a low-loss medium, the effective per-unit-length capacitance and inductance can be reverse-calculated directly from the CST outputs:

$$Z_0 = \sqrt{\frac{L}{C}} \implies L = Z_0^2 \cdot C \quad (5)$$

$$\beta = \omega\sqrt{LC} = \omega\sqrt{Z_0^2 \cdot C \cdot C} = \omega Z_0 C \implies C = \frac{\beta}{\omega Z_0} \quad (6)$$

Substituting the CST simulated values ($Z_0 = 69.2213 \Omega$, $\beta = 15.18 \text{ rad/m}$) at $\omega = 2\pi \times 10^9 \text{ rad/s}$:

$$C = \frac{15.18}{2\pi \times 10^9 \times 69.2213} = 3.49 \times 10^{-11} \text{ F/m} \quad (7)$$

$$L = (69.2213)^2 \times (3.49 \times 10^{-11}) = 1.67 \times 10^{-7} \text{ H/m} \quad (8)$$

These derived effective values account for the 3D modal distributions and meshing approximations present in the full-wave numerical solver, providing a robust comparison against the idealized 1D theoretical model.